

Projected Gradient Descent: Algorithm and Convergence Analysis

1 Algorithm Description

The algorithm iteratively updates parameters by taking gradient steps and projecting the result back onto the constraint set at each iteration. It is designed for smooth convex optimization problems with closed, convex, and bounded constraint domains.

1.1 Algorithm Name

Projected Gradient Descent (PGD)

1.2 Mathematical Formulation

Given a smooth convex objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and a closed, convex, and bounded constraint set $C \subseteq \mathbb{R}^n$, the algorithm solves the optimization problem:

$$\min_{w \in C} f(w).$$

The update rule at iteration t is:

$$w_{t+1} = \Pi_C (w_t - \eta_t \nabla f(w_t)),$$

where:

- $\nabla f(w_t)$ is the gradient of f at w_t ,
- $\eta_t > 0$ is the step size (learning rate),
- $\Pi_C(\cdot)$ is the Euclidean projection operator onto C , defined as:

$$\Pi_C(v) = \arg \min_{w \in C} \|w - v\|_2^2,$$

where $\|\cdot\|_2$ is the standard Euclidean norm.

Hyperparameters:

- **Step size** η_t : Controls the magnitude of parameter updates. Can be fixed ($\eta_t = \eta$) or adaptively adjusted.
- **Initialization** w_0 : Initial parameter vector, typically set to 0 or random values within C .
- **Termination criterion**: Iteration stops when $\|w_{t+1} - w_t\|_2 < \epsilon$ (for small $\epsilon > 0$) or a maximum number of iterations T is reached.

2 Pseudo Code and Dry Run

2.1 Pseudocode

Inputs:

Algorithm 1 Projected Gradient Descent

```
1: procedure PROJECTEDGRADIENTDESCENT( $f, \text{grad}_f, C, w_0, \eta, \text{max\_iter}, \epsilon$ )
2:    $w \leftarrow w_0$ 
3:   for  $t = 0$  to  $\text{max\_iter} - 1$  do
4:      $\text{gradient} \leftarrow \text{grad}_f(w)$  ▷ Gradient computation
5:      $\tilde{w} \leftarrow w - \eta \cdot \text{gradient}$  ▷ Gradient step
6:      $w_{\text{new}} \leftarrow \Pi_C(\tilde{w})$  ▷ Projection onto  $C$ 
7:     if  $\|w_{\text{new}} - w\|_2 < \epsilon$  then ▷ Termination condition
8:       break
9:     end if
10:     $w \leftarrow w_{\text{new}}$ 
11:  end for
12:  return  $w$ 
13: end procedure
```

- f : Objective function.
- grad_f : Gradient of f .
- C : Constraint set with a projection operator Π_C .
- w_0 : Initial parameter vector.
- η : Fixed step size.
- max_iter : Maximum iterations.
- ϵ : Convergence tolerance.

Outputs:

- w : Final parameter vector.

2.2 Dry Run Example

Objective Function: $f(w) = (w - 3)^2$, with gradient $\nabla f(w) = 2(w - 3)$. **Constraint Set:** $C = \mathbb{R}$ (unbounded, but valid for demonstration). Projection Π_C is identity. **Hyperparameters:** $w_0 = 0$, $\eta = 0.1$, minibatch size = 1, $\epsilon = 10^{-4}$.

2.2.1 Iteration 1

1. **Gradient computation:** $\nabla f(w_0) = 2(0 - 3) = -6$.
2. **Parameter update:** $\tilde{w}_1 = 0 - 0.1 \cdot (-6) = 0.6$. Projection: $w_1 = \Pi_C(0.6) = 0.6$.
3. **Objective value:** $f(w_1) = (0.6 - 3)^2 = (-2.4)^2 = 5.76$.

2.2.2 Iteration 2

1. **Gradient computation:** $\nabla f(w_1) = 2(0.6 - 3) = -4.8$.
2. **Parameter update:** $\tilde{w}_2 = 0.6 - 0.1 \cdot (-4.8) = 1.08$. Projection: $w_2 = \Pi_C(1.08) = 1.08$.
3. **Objective value:** $f(w_2) = (1.08 - 3)^2 = (-1.92)^2 = 3.6864$.

2.2.3 Iteration 3

1. **Gradient computation:** $\nabla f(w_2) = 2(1.08 - 3) = -3.84$.
2. **Parameter update:** $\tilde{w}_3 = 1.08 - 0.1 \cdot (-3.84) = 1.464$. Projection: $w_3 = \Pi_C(1.464) = 1.464$.
3. **Objective value:** $f(w_3) = (1.464 - 3)^2 = (-1.536)^2 = 2.359296$.

Summary:

Iteration t	w_t	$\nabla f(w_t)$	$f(w_t)$
0	0	-6	9.0
1	0.6	-4.8	5.76
2	1.08	-3.84	3.6864
3	1.464	-3.072	2.359296

Table 1: Summary of iterations

The algorithm reduces $f(w)$ monotonically, approaching the unconstrained minimum at $w = 3$.

3 Convergence Theory

3.1 Assumptions

We analyze the convergence of the Projected Gradient Descent (PGD) algorithm under the following assumptions:

Assumption 1 (Convexity and Smoothness). *The objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and twice continuously differentiable. Its gradient is L -Lipschitz continuous, meaning:*

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \forall x, y \in \mathbb{R}^n.$$

This ensures f satisfies the descent lemma:

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2.$$

Assumption 2 (Constraint Set). *The constraint set $C \subset \mathbb{R}^n$ is:*

- **Closed and convex**, ensuring the projection operator P_C is well-defined and non-expansive.
- **Bounded**, with a finite diameter $D = \sup_{x, y \in C} \|x - y\|$.

Assumption 3 (Step Size). *The step size satisfies $\eta_k = 1/L$ for all iterations k .*

These assumptions guarantee that PGD iterates remain within C while leveraging smoothness and convexity for convergence.

3.2 Main Convergence Theorem

We state the convergence properties of PGD under the above assumptions.

Theorem 1 (Sublinear Convergence Rate). *Let f and C satisfy Assumptions 1-2, and let $x^* \in \arg \min_{x \in C} f(x)$. For iterates $\{x_k\}_{k=0}^{T-1}$ generated by PGD with $\eta_k = 1/L$, define the average iterate $\bar{x}_T = \frac{1}{T} \sum_{k=0}^{T-1} x_k$. Then:*

$$f(\bar{x}_T) - f(x^*) \leq \frac{L\|x_0 - x^*\|^2}{2T}.$$

Proof. By the descent lemma and projection inequality, we derive the key bound:

$$2\eta(f(x_k) - f(x^*)) \leq \|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2.$$

Summing over $k = 0$ to $T - 1$, telescoping yields:

$$2\eta \sum_{k=0}^{T-1} (f(x_k) - f(x^*)) \leq \|x_0 - x^*\|^2.$$

Dividing by $2\eta T$ and applying convexity ($f(\bar{x}_T) \leq \frac{1}{T} \sum f(x_k)$) gives the result. $\square$

Corollary 1 (Special Case: Strong Convexity). *If f is μ -strongly convex, PGD achieves linear convergence:*

$$f(x_k) - f(x^*) \leq \frac{L}{2}(1 - \mu/L)^k \|x_0 - x^*\|^2.$$

3.3 Theoretical Properties

3.3.1 Stability Analysis

The projection operator P_C is non-expansive ($\|P_C(x) - P_C(y)\| \leq \|x - y\|$), ensuring iterates remain within C . This guarantees numerical stability and robustness to perturbations in gradient estimates.

3.3.2 Sensitivity to Hyperparameters

The step size $\eta = 1/L$ is critical. If $\eta > 1/L$, the descent lemma fails, leading to divergence. If L is unknown, adaptive methods (e.g., backtracking line search) can stabilize convergence at the cost of additional gradient evaluations.

3.3.3 Comparison to Baseline Methods

- **Frank-Wolfe Algorithm:** Avoids projections but requires linear optimization over C , with similar $O(1/T)$ rates for smooth convex objectives.
- **Accelerated PGD:** Achieves $O(1/T^2)$ rates for smooth convex functions but requires careful momentum coefficient updates.
- **Interior-Point Methods:** Faster local convergence for structured problems but incompatible with high-dimensional or unstructured constraints.

3.4 Discussion

The $O(1/T)$ rate in Theorem 1 matches lower bounds for smooth convex minimization without acceleration. Boundedness of C (Assumption 2) ensures finite diameter D , critical for the Lyapunov analysis. While PGD is slower than accelerated methods, its simplicity and versatility make it robust for large-scale or nonsmooth extensions (e.g., proximal gradient methods). Future work could explore adaptive restarts or stochastic variants for scalability.

4 Convergence Proofs

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

Convex Function: A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if for all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Smooth Function: A function f is L -smooth if its gradient is L -Lipschitz continuous:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \forall x, y \in \mathbb{R}^n.$$

Projection Operator: For a closed convex set $C \subset \mathbb{R}^n$, the projection $P_C(x)$ is the unique point in C satisfying:

$$\|x - P_C(x)\| \leq \|x - y\| \quad \forall y \in C.$$

4.1.2 Lemmas

Descent Lemma: For an L -smooth function f , the following holds for all $x, y \in \mathbb{R}^n$:

$$f(y) \leq f(x) + \nabla f(x)^T(y - x) + \frac{L}{2}\|y - x\|^2.$$

Convexity Inequality: For a convex function f , the gradient satisfies:

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) \quad \forall x, y \in \mathbb{R}^n.$$

Projection Inequality: For any $z \in C$, the projection $x_{k+1} = P_C(x_k - \eta \nabla f(x_k))$ satisfies:

$$(x_{k+1} - x_k + \eta \nabla f(x_k))^T(x_{k+1} - z) \leq 0.$$

Non-Expansiveness of Projection: For any $x, y \in \mathbb{R}^n$:

$$\|P_C(x) - P_C(y)\| \leq \|x - y\|.$$

4.2 Main Proof (Step-by-Step Derivation)

4.2.1 Key Inequality Derivation

We aim to bound $\|x_{k+1} - x^*\|^2$ in terms of $\|x_k - x^*\|^2$ and $f(x_k) - f(x^*)$. Start with the projection inequality:

$$(x_{k+1} - x_k + \eta \nabla f(x_k))^T(x_{k+1} - x^*) \leq 0.$$

Expanding this:

$$(x_{k+1} - x_k)^T(x_{k+1} - x^*) + \eta \nabla f(x_k)^T(x_{k+1} - x^*) \leq 0.$$

Rewriting the first term:

$$\|x_{k+1} - x_k\|^2 + (x_k - x^*)^T(x_{k+1} - x_k) + \eta \nabla f(x_k)^T(x_{k+1} - x^*) \leq 0.$$

Now, expand $\|x_{k+1} - x^*\|^2$:

$$\|x_{k+1} - x^*\|^2 = \|x_{k+1} - x_k + x_k - x^*\|^2 = \|x_{k+1} - x_k\|^2 + \|x_k - x^*\|^2 + 2(x_{k+1} - x_k)^T(x_k - x^*).$$

Substitute the earlier inequality into this expansion:

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - \|x_{k+1} - x_k\|^2 - 2\eta \nabla f(x_k)^T(x_{k+1} - x^*).$$

Apply convexity:

$$\nabla f(x_k)^T(x_{k+1} - x^*) = \nabla f(x_k)^T(x_{k+1} - x_k + x_k - x^*) \geq f(x_{k+1}) - f(x_k) + f(x_k) - f(x^*) = f(x_{k+1}) - f(x^*).$$

Substitute into the inequality:

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - 2\eta(f(x_{k+1}) - f(x^*)) - \|x_{k+1} - x_k\|^2.$$

However, using the projection condition again, we derive a tighter bound. From the projection inequality:

$$\|x_{k+1} - x_k\|^2 + 2\eta \nabla f(x_k)^T(x_{k+1} - x_k) \leq 0.$$

This implies:

$$-2\eta \nabla f(x_k)^T (x_{k+1} - x_k) \geq \|x_{k+1} - x_k\|^2.$$

Returning to the expanded $\|x_{k+1} - x^*\|^2$:

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - \|x_{k+1} - x_k\|^2 - 2\eta \nabla f(x_k)^T (x_{k+1} - x_k) - 2\eta \nabla f(x_k)^T (x_k - x^*).$$

Substitute the bound on $-2\eta \nabla f(x_k)^T (x_{k+1} - x_k)$:

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - 2\eta \nabla f(x_k)^T (x_k - x^*).$$

Apply convexity:

$$\nabla f(x_k)^T (x_k - x^*) \geq f(x_k) - f(x^*).$$

Thus:

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - 2\eta (f(x_k) - f(x^*)).$$

4.2.2 Telescoping Sum and Convexity

Summing over $k = 0$ to $T - 1$:

$$\sum_{k=0}^{T-1} (\|x_{k+1} - x^*\|^2 - \|x_k - x^*\|^2) \leq -2\eta \sum_{k=0}^{T-1} (f(x_k) - f(x^*)).$$

The left-hand side telescopes:

$$\|x_T - x^*\|^2 - \|x_0 - x^*\|^2 \leq -2\eta \sum_{k=0}^{T-1} (f(x_k) - f(x^*)).$$

Rearranging:

$$2\eta \sum_{k=0}^{T-1} (f(x_k) - f(x^*)) \leq \|x_0 - x^*\|^2.$$

Divide by $2\eta T$:

$$\frac{1}{T} \sum_{k=0}^{T-1} (f(x_k) - f(x^*)) \leq \frac{\|x_0 - x^*\|^2}{2\eta T}.$$

By convexity of f , $f(\bar{x}_T) \leq \frac{1}{T} \sum_{k=0}^{T-1} f(x_k)$. Therefore:

$$f(\bar{x}_T) - f(x^*) \leq \frac{\|x_0 - x^*\|^2}{2\eta T}.$$

With $\eta = 1/L$, this becomes:

$$f(\bar{x}_T) - f(x^*) \leq \frac{L\|x_0 - x^*\|^2}{2T}.$$

4.3 Rate Derivation (With Constants)

The derived inequality shows that the average function value $f(\bar{x}_T)$ converges to $f(x^*)$ at a rate of $O(1/T)$. The constant depends on the initial distance to the optimal solution $\|x_0 - x^*\|^2$ and the smoothness parameter L . This sublinear rate is optimal for gradient descent on smooth convex functions without additional assumptions.

4.4 Special Cases

4.4.1 Strong Convexity

If f is μ -strongly convex, then:

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) + \frac{\mu}{2}\|y - x\|^2 \quad \forall x, y \in \mathbb{R}^n.$$

Combining this with the key inequality:

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - 2\eta(f(x_k) - f(x^*)) \leq \|x_k - x^*\|^2 - \eta\mu\|x_k - x^*\|^2.$$

Thus:

$$\|x_{k+1} - x^*\|^2 \leq (1 - \eta\mu)\|x_k - x^*\|^2.$$

With $\eta = 1/L$, this becomes:

$$\|x_k - x^*\|^2 \leq \left(1 - \frac{\mu}{L}\right)^k \|x_0 - x^*\|^2.$$

Using smoothness:

$$f(x_k) - f(x^*) \leq \frac{L}{2}\|x_k - x^*\|^2 \leq \frac{L}{2} \left(1 - \frac{\mu}{L}\right)^k \|x_0 - x^*\|^2.$$

This establishes a linear convergence rate for strongly convex functions.

4.5 Summary

The convergence analysis of Projected Gradient Descent under smoothness and convexity assumptions yields a sublinear $O(1/T)$ rate for the average iterate. Strong convexity improves this to a linear rate. The proof relies on the descent lemma, convexity inequalities, and projection properties, ensuring all steps are rigorously justified.